

Name: _____ Date: _____

Period: _____

1. A student needs to find the **vertical asymptotes and holes** of a rational function. Circle the form that is most useful, and explain why in one phrase.

A. Factored form**B. Standard form**

Because: **factored form shows the zeros of the numerator and denominator, making it easy to identify holes and vertical asymptotes by comparing shared factors.**.....

2. Perform the first step of polynomial long division to divide $x^2 + 3x + 5$ by $x + 2$.

What is the first term of the quotient? x

After multiplying and subtracting, what is the new dividend? $x + 5$

3. Use Pascal's Triangle to expand $(x + 2)^4$. (Row 4 of Pascal's Triangle: 1 4 6 4 1)

$$(x + 2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$$

Teacher Note

Item 2: First term of quotient $= x^2 \div x = x$. Multiply: $x(x + 2) = x^2 + 2x$. Subtract from $x^2 + 3x + 5$: new dividend is $x + 5$. Final answer (if completed): quotient $x + 1$, remainder 3, so $\frac{x^2+3x+5}{x+2} = x + 1 + \frac{3}{x+2}$.

Item 3: $(x + 2)^4 = 1 \cdot x^4 + 4 \cdot x^3 \cdot 2 + 6 \cdot x^2 \cdot 4 + 4 \cdot x \cdot 8 + 1 \cdot 16 = x^4 + 8x^3 + 24x^2 + 32x + 16$.